Stridr - Functional Design Document

Version: 1.0
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Status: Pre-Launch

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1. Executive Summary

Stridr is a mobile fitness gamification application that transforms daily walking into virtual trekking adventures. Users track their real-world steps to virtually progress through iconic Indian Himalayan trails, unlocking landmarks, earning badges, and building healthy walking habits.

Target Platforms

- **iOS** (iPhone/iPad)
- **Android**

Core Value Proposition

Turn mundane daily walks into exciting virtual adventures exploring India's most beautiful trekking routes.

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2. Application Architecture

2.1 Technology Stack

Component	Technology
Framework	Expo SDK 54 / React Native 0.81
Language	TypeScript
Navigation	Expo Router (file-based)
State Management	React Context API
Local Storage	AsyncStorage
Sensors	expo-sensors (Pedometer)
Notifications	expo-notifications
Background Tasks	expo-background-fetch, expo-task-manager
UI Icons	lucide-react-native

2.2 Application Structure

```
├── app/                    # Screen components (Expo Router)
│   ├── (tabs)/             # Tab navigation screens
│   │   ├── index.tsx       # Home screen
│   │   ├── trails.tsx      # Trail selection
│   │   ├── progress.tsx    # Active trail progress
│   │   ├── achievements.tsx# Badge collection
```

```
│   │   └── profile.tsx     # User profile & settings
│   ├── trail/[id].tsx      # Trail detail screen
│   ├── login.tsx           # User authentication
│   ├── signup.tsx          # User registration
│   ├── my-dashboard.tsx    # User statistics dashboard
│   ├── edit-profile.tsx    # Profile editing
│   ├── notification-settings.tsx
│   └── legal.tsx           # Privacy policy & ToS
├── src/
│   ├── context/            # React Context providers
│   ├── services/           # Business logic services
│   ├── const/              # Static data (trails, badges)
│   ├── types/              # TypeScript interfaces
│   └── utils/              # Utility functions
```

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3. Core Features

3.1 User Authentication

Purpose: Secure user account management with local-first data storage.

Feature	Description
Registration	First name, last name, email, password with validation
Login	Email/password authentication
Session Persistence	Auto-login via stored session token
Profile Updates	Edit name, email, profile photo
Account Deletion	GDPR-compliant full data removal

> *[!NOTE] Social authentication (Google/Facebook) deferred for initial launch; requires development build for production implementation.*

3.2 Step Tracking System

Purpose: Track user's physical activity and convert steps to virtual trail distance.

How It Works

1. **Pedometer Integration** - Uses device's built-in motion sensors via `expo-sensors`
2. **Step Calculation** - Converts steps to distance using configurable stride length (default: 75cm)
3. **Sync Mechanism** - Automatically fetches steps since last sync
4. **Daily Logging** - Records daily step counts for historical tracking

Data Model

```typescript
interface UserProgress {
  selectedTrailId: string | null;
  trailStartDate: string | null;
  targetDays: number;
```

```
  totalStepsValid: number;
  currentDistanceMeters: number;
  lastSyncTime: string;
  unlockedBadges: string[];
  completedTrails: CompletedTrail[];
  favoriteTrails: string[];
  currentStreak: number;
  lastLogDate: string | null;
}
```

3.3 Virtual Trails

Purpose: Provide engaging virtual trekking experiences mapped to real Indian Himalayan trails.

Available Trails (Incredible India Collection)

Trail Name	Distance	Difficulty	Landmarks
Roopkund Trek	53 km	Hard	8 landmarks
Chadar Trek	65 km	Extreme	6 landmarks
Kedarkantha Trek	20 km	Easy	6 landmarks
Valley of Flowers	38 km	Moderate	6 landmarks
Hampta Pass Trek	26 km	Moderate	6 landmarks
Goechala Trek	90 km	Hard	7 landmarks
Markha Valley Trek	78 km	Hard	6 landmarks
Pin Parvati Pass	110 km	Extreme	8 landmarks
Sandakphu Trek	64 km	Moderate	6 landmarks
Stok Kangri	16 km	Extreme	5 landmarks

Trail Data Structure

```
interface Trail {
  id: string;
  name: string;
  description: string;
  extendedDescription?: string;
  totalDistanceMeters: number;
  image: any;
  color: string;
  difficulty: 'Easy' | 'Moderate' | 'Hard' | 'Extreme';
  landmarks: Landmark[];
  startCoordinate?: { latitude: number; longitude: number };
  endCoordinate?: { latitude: number; longitude: number };
  region?: { ... };
}
```

```
interface Landmark {
  id: string;
  name: string;
  distanceMeters: number;
  description: string;
  image?: any;
  coordinate?: { latitude: number; longitude: number };
}
```

3.4 Gamification System

3.4.1 Badge Collections

Six badge collections rewarding different achievements:

Collection	Focus	Badges
🚶 Walking Warrior	Total steps accumulated	9 badges (1K to 1M steps)
🗺️ Distance Destroyer	Total distance covered	10 badges (1km to 1000km)
🔥 Streak Star	Consecutive walking days	Multiple streak tiers
⛰️ Trail Blazer	Trail completions	Per-trail completion
🌍 Seasonal Wanderers	Seasonal achievements	Weather/season themed
⭐ Legendary Life	Special achievements	Rare accomplishments

Badge Unlock Conditions

```
interface Badge {
  id: string;
  name: string;
  description: string;
  icon: string; // Emoji
  conditionType: 'TOTAL_STEPS' | 'TOTAL_DISTANCE' | 'STREAK' | 'TRAIL_COMPLETE' | 'SEASON' |
'MONTH';
  conditionValue: number;
  collection: string;
}
```

3.4.2 Progress Tracking

- **Daily Goal** - Default 10,000 steps (configurable)
- **Weekly Activity Chart** - Visual 7-day step history
- **Trail Progress** - Percentage completion with landmark milestones
- **Streak System** - Consecutive days of meeting daily goal

3.5 Push Notifications

Purpose: Keep users engaged and motivated with timely notifications.

Notification Type	When Triggered
Daily Reminder	Scheduled time (morning/afternoon/evening)
Goal Achievement	User reaches daily step goal
Badge Unlock	New badge earned
Milestone	25%, 50%, 75% trail completion
Landmark Reached	User reaches a trail landmark
Inactivity Nudge	No activity for 2+ days

Notification Settings Model

```
interface NotificationSettings {
  dailyReminder: boolean;
  dailyReminderTime: 'morning' | 'afternoon' | 'evening';
  goalAchievement: boolean;
  badgeUnlock: boolean;
  milestone: boolean;
  inactivityNudge: boolean;
  landmarkReached: boolean;
}
```

3.6 User Dashboard

Purpose: Centralized view of personal statistics and achievements.

Dashboard Components

- **All-Time Statistics**
  - Total steps
  - Total distance (km/mi)
  - Total days active
  - Trails completed
  - Badges earned
  - Longest streak
- **Completed Trail Cards**
  - Trail image background
  - Completion date
  - Total steps/days
  - Average steps per day
  - Max steps in one day

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4. Screen Specifications

4.1 Home Screen (`index.tsx`)

Primary Purpose: Dashboard overview and quick access to current activity.

Section	Content
Header	Greeting with user name
Daily Goal Card	Circular progress, today's steps vs goal
Active Trail	Current trail progress with image, distance remaining
Weekly Activity	7-day bar chart of steps
Your Accomplishments	Badge count, days active, total distance
Quick Start	Suggested trails for new users

4.2 Trails Screen (`trails.tsx`)

Primary Purpose: Browse and select virtual trails.

Section	Content
Trail List	Cards with name, difficulty, distance, landmarks
Favorites	Heart toggle to mark favorite trails
Coming Soon	Banner for upcoming trail additions

4.3 Progress Screen (`progress.tsx`)

Primary Purpose: Detailed view of active trail progress.

Section	Content
Trail Banner	Hero image with trail name overlay
Progress Bar	Visual completion percentage
Stats Row	Steps, distance, days on trail
Trail Activity Chart	Visual representation of progress over time
Landmarks Timeline	Vertical timeline showing unlocked/locked landmarks

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4.4 Achievements Screen (`achievements.tsx`)

Primary Purpose: Badge collection and gamification rewards.

Section	Content
Summary Stats	Total badges earned / available
Collections	Grouped badges by category
Badge Cards	Icon, name, description, unlock status

4.5 Profile Screen (`profile.tsx`)

Primary Purpose: User settings and account management.

Section	Content
Profile Header	Photo, name, edit button
Favorite Trails	Quick access to saved trails
Dashboard Link	Navigate to full statistics
Settings	Units, notifications, stride length
Legal	Privacy policy, terms of service
Account Actions	Logout, delete account

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5. User Preferences

5.1 Configurable Settings

Setting	Options	Default
Distance Unit	Kilometers / Miles	km
Theme	Light / Dark	Light
Daily Goal	Any number	10,000 steps
Stride Length	Configurable (cm)	75 cm
Notifications	Toggles per type	All enabled
Reminder Time	Morning/Afternoon/Evening	Morning (9 AM)

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6. Data Persistence

6.1 Local Storage Keys

Key	Purpose
`stridr_users_db`	User accounts database
`stridr_current_session`	Active user session
`user_progress_{userId}`	User's trail progress
`daily_logs_{userId}`	Historical step data
`preferences`	User settings

6.2 Data Flow

```
graph TD
    A[Device Pedometer] -->|Steps| B[StepService]
    B --> C[GameContext]
    C --> D[StorageService]
    D --> E[AsyncStorage]
    C --> F[BadgeService]
    C --> G[StatsService]
    F --> H[NotificationService]
```

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7. Service Layer

7.1 Service Overview

Service	Responsibility
`StepService`	Fetch steps from device pedometer
`StorageService`	Persist/retrieve data from AsyncStorage
`BadgeService`	Check badge unlock conditions
`StatsService`	Calculate statistics (averages, totals, streaks)
`NotificationService`	Schedule and send push notifications

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8. Security & Privacy

8.1 Data Handling

- All user data stored locally on device (AsyncStorage)
- No cloud backend – fully offline capable
- Passwords stored in local DB (simple implementation for MVP)
- GDPR-compliant: Delete Account removes all data
- Motion data only used for step counting (disclosed in app)

8.2 Required Permissions

Permission	Platform	Purpose
Motion & Fitness	iOS	Step counting
Photo Library	iOS	Profile photo upload
Camera	iOS	Taking profile photos
Notifications	Both	Push notification delivery
Boot Completed	Android	Background task restoration

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9. Future Enhancements

9.1 Planned Features

Feature	Priority	Notes
Social Login	High	Google/Facebook OAuth with dev build
Cloud Sync	Medium	Backup progress across devices
Social Features	Medium	Friend challenges, leaderboards
More Trail Regions	Medium	World trails (Americas, Europe, Asia)
Apple Health Integration	Low	Sync with Apple Health data
Google Fit Integration	Low	Sync with Google Fit data

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10. Appendix

10.1 File Reference

Path	Description
`app.json`	Expo configuration, permissions, bundle IDs
`src/const/trails.ts`	All trail definitions
`src/const/badges.ts`	Badge collection definitions
`src/context/GameContext.tsx`	Core game state management
`src/context/AuthContext.tsx`	Authentication state
`src/context/PreferencesContext.tsx`	User preferences state

10.2 Bundle Identifiers

- **iOS:** `com.newrishav.stridr`
- **Android:** `com.newrishav.stridr`

10.3 App Store Category

- **Primary:** Health & Fitness
- **Age Rating:** 4+

This document describes the functional design of Stridr v1.0.0